SPACE ROBOTICS LABORATORY · TOHOKU UNIVERSITY · PHASE 0

# Moonwalk 2.0

## Report 04 · From the plan to the robot

Report 03 grew the block into structures. This report asks how they get built: where, in what order, and by which robot moves. It checks the site against ispace's Mission 4 landing zone, turns the pad and the dome into assembly plans the Dragon can execute, and sets out how they are simulated



TOWARDS A PERMANENT [illegible] SINCE ON THE MOON

The SPACE ROBOTICS LAB

SPACE ROBOTICS LABORATORY
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CONFERENCES AND DEADLINES

LIVE REPORT
arhimatrix.github.io/arhimatrix-moonwalk-2-rep_04

INTERACTIVE PARTS
3D Moon tour with the 87° line · 3D build players for the pad and the dome · videos · job lists (CSV, JSON)

01

SHACKLETON–DE GERLACHE RIDGE · LROC · NASA/GSFC/ARIZONA STATE UNIVERSITY



CHAPTER 01 · MOONWALK 2.0 · REPORT 04

# The site

## Connecting Ridge, between Shackleton and de Gerlache

Report 02 scored four NASA candidate regions; the pad and dome in this report are planned for Connecting Ridge: long sunlit periods, a shadowed floor within reach, and a shoulder flat enough for the pad.

01 · ORIENTATION

# Turning to the south pole

The Moon's axis is tilted only about 1.5°. At the pole the Sun never climbs more than a few degrees: light arrives sideways, high ground can stay lit for most of the year, and low ground can stay dark for ever.



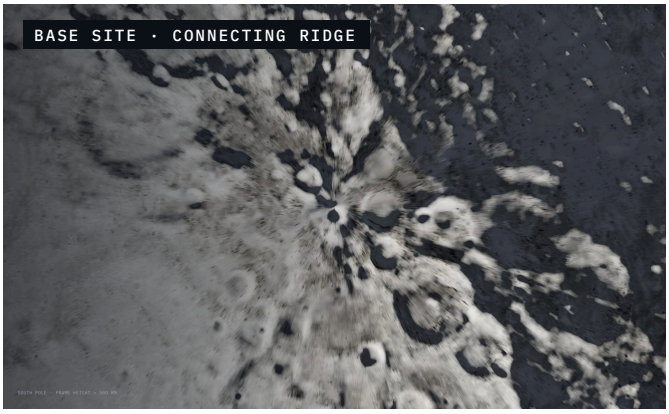

THE 3D MOON OF THE LIVE REPORT TURNS TO THE SOUTH POLE

01 · THE SITE

## Connecting Ridge

Within 150 km of the pole the crater rims catch the low Sun while their floors never see it. The pad and dome in this report are planned for Connecting Ridge, between Shackleton and de Gerlache.

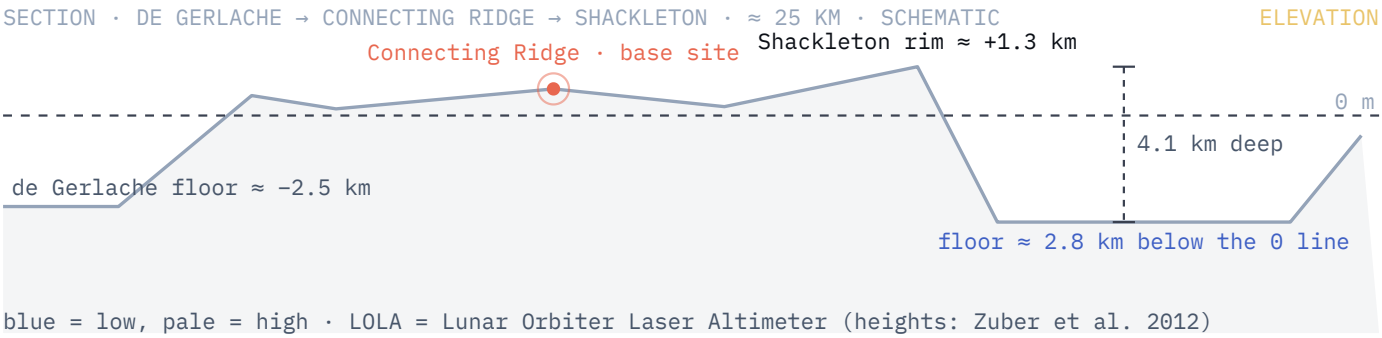
The region opens in JSMARS, a browser-based planetary GIS by Prof. Anwar Sounny-Slitine: jsmars.sounny.com, lat −89.46°, lon −99°.



01 · SITE DATA

# Height, light and cold

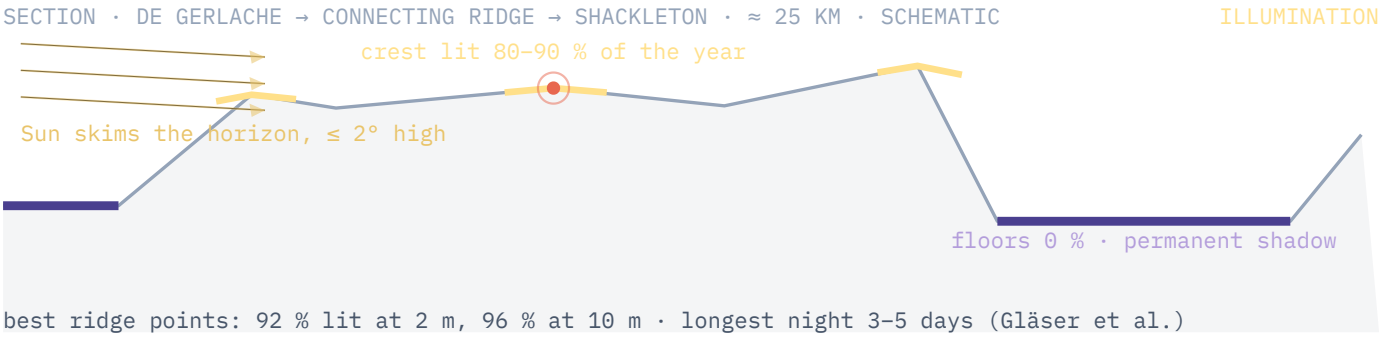
Three NASA data sets explain the choice of the ridge, drawn on one schematic section from de Gerlache across Connecting Ridge to Shackleton.




**Elevation · LOLA**

The base site sits on the high crest between the two craters, because at the pole height buys sunlight.

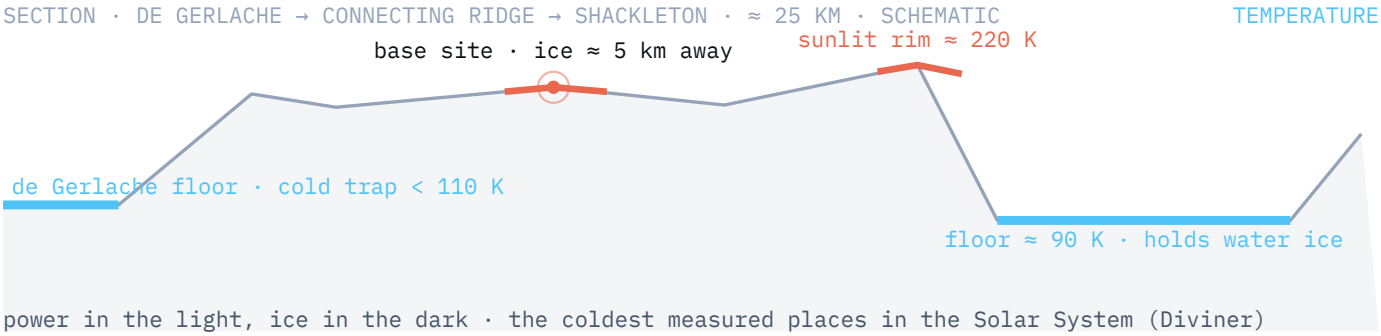
Report layer: LOLA 5 m polar DEM (Zuber et al. 2012).




**Illumination**

The crest is lit 80–90 % of the year; the best ridge points 92 % at 2 m and 96 % at 10 m, with the longest night 3–5 days.

Report layer: Mazarico average-illumination map; values after Gläser et al.




**Temperature · Diviner**

Floors that never see the Sun stay below 110 K; Shackleton's floor, at about 90 K, holds water ice about 5 km from the base site.

Report layer: Diviner maximum-temperature map.

01 · SITE DATA ON THE MOON

# The same three layers over the region

On the 3D Moon of the live report each data set is laid over the polar region, with the base site on Connecting Ridge marked. Schematic layers; the report versions use the NASA maps named in the captions.




ELEVATION · LOLA POLAR ELEVATION: BLUE LOW, PALE HIGH




ILLUMINATION, % OF THE YEAR: THE CREST EXCEEDS 80 %




DIVINER COLD TRAPS: FLOORS BELOW 110 K




CHAPTER 02 · MOONWALK 2.0 · REPORT 04

# Landing zone

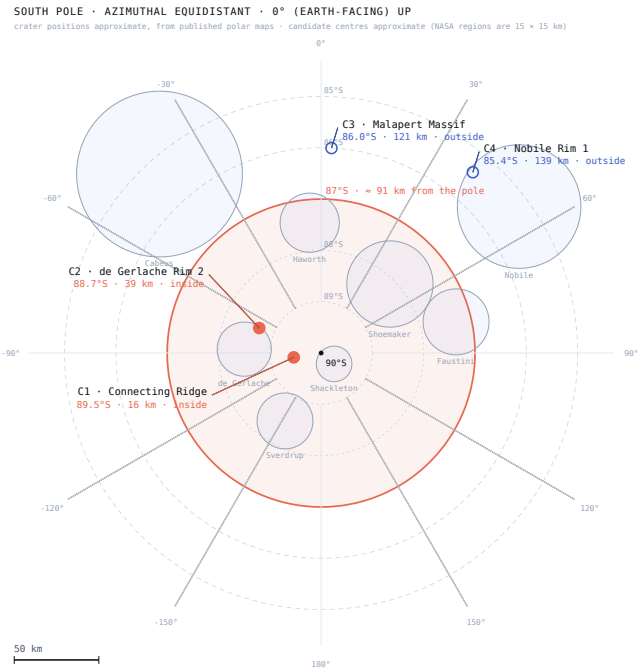
Do the candidate sites fall inside ispace's Mission 4 zone?

The Mission 4 landing latitude given is 87° South or higher: a circle about 91 km in radius around the pole. Connecting Ridge and de Gerlache Rim 2 are inside it; Malapert Massif and Nobile Rim 1 are outside.

02 · LANDING ZONE

# Do the candidate sites fall inside ispace's Mission 4 zone?

ispace has set the Mission 4 landing latitude at 87° South or higher. Drawn on the site map, that line is a circle about 182 km across around the pole. Two of the four candidate regions lie inside it, including Connecting Ridge, the base site of these reports.




SOUTH POLE, AZIMUTHAL EQUIDISTANT PROJECTION, 0° (THE EARTH-FACING SIDE) AT THE TOP. ORANGE: THE ≥ 87°S BAND. FILLED DOTS: CANDIDATES INSIDE IT; OPEN CIRCLES: OUTSIDE.

| CANDIDATE | LATITUDE | FROM THE POLE | ≥ 87°S |
|---|---|---|---|
| **C1 · Connecting Ridge** BASE SITE | 89.5°S | 16 km | INSIDE |
| **C2 · de Gerlache Rim 2** | 88.7°S | 39 km | INSIDE |
| **C3 · Malapert Massif** | 86.0°S | 121 km | OUTSIDE |
| **C4 · Nobile Rim 1** | 85.4°S | 139 km | OUTSIDE |




THE ≥ 87°S CIRCLE ON THE 3D MOON OF THE LIVE REPORT, WITH THE FOUR CANDIDATES LABELLED

WHAT ISPACE HAS PUBLISHED

## Mission 4 · 2029 · ULTRA lander

Mission 4 (renumbered from Mission 6 in March 2026) is planned for 2029 on ispace's larger ULTRA lander, about 3.6 m tall and about 1,000 kg dry. It is funded under Japan's Space Strategy Fund, "high-precision landing technology in the lunar poles". The aims are a landing near the south pole with an accuracy of 100 m or better (at least SLIM level) and more than 14 days of operation, using the low Sun on the lander's sides. The public documents name the south polar region but give no latitude. The 87° figure is the one given to us, and it is drawn here as stated.

A WIDER REGION THAN NASA'S

## A band of about 26,000 km²

NASA names 15 × 15 km regions for crewed landings. A ≥ 87°S zone is a band of about 26,000 km² around the pole. The band says where the lander may go; the 100 m accuracy is what lets it land next to a chosen pad site inside that band.

WHAT IT MEANS FOR MOONWALK

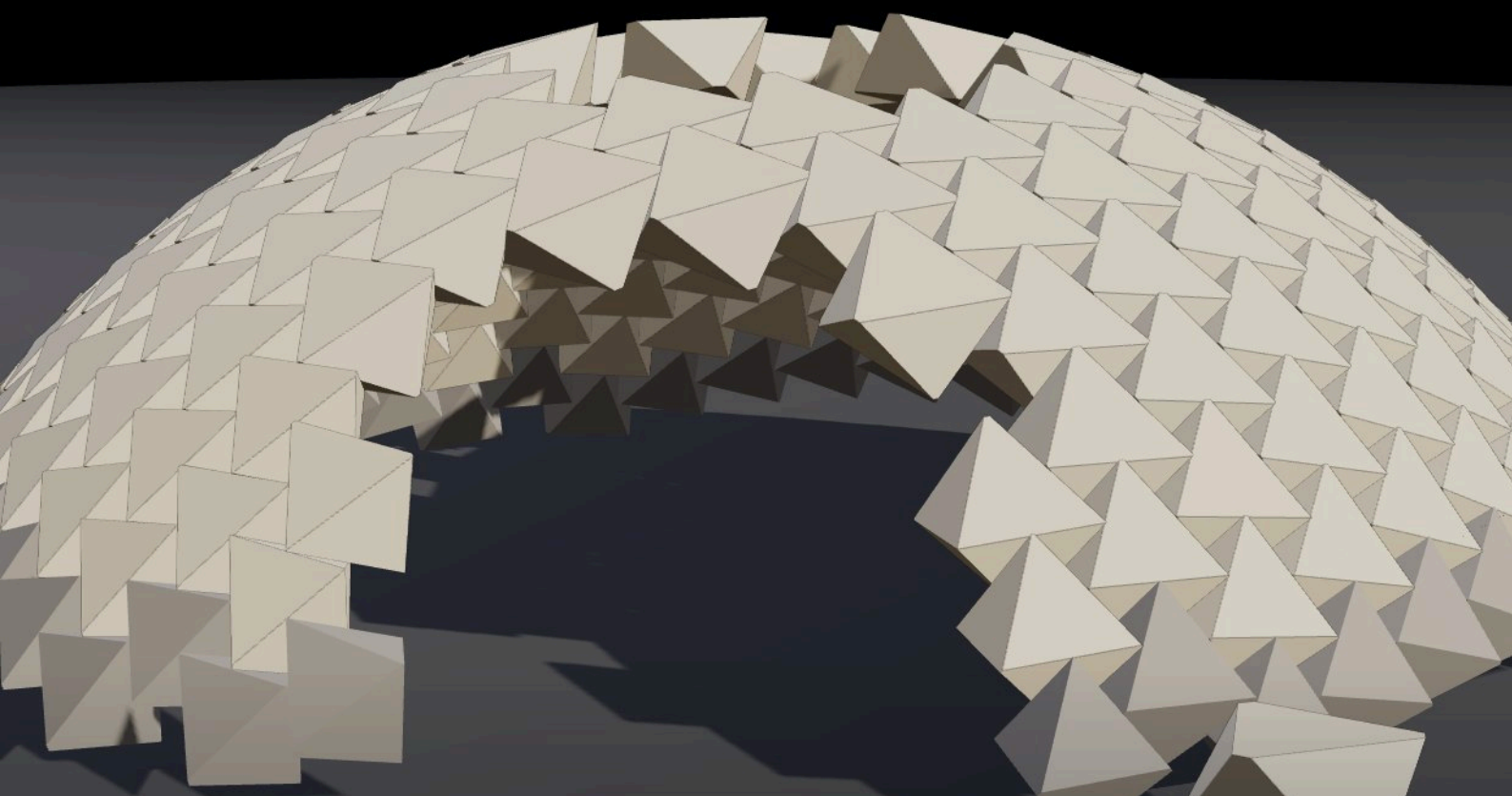
## Two candidates qualify

Connecting Ridge and de Gerlache Rim 2 both qualify. Malapert, the best site for the Earth link, and Nobile Rim 1 do not. The pad and dome below are therefore planned for a site a Mission-4-class lander could serve. The pad sits outside the landing ellipse, and the robot's first job is the drive from the lander to the pad site.

SOURCES

ispace, Mission 4 (ispace-inc.com/mission-4) · ispace, mission timeline update, 30 March 2026 · ispace, Space Strategy Fund grant, 7 August 2026 · MEXT, Space Strategy Fund, lunar polar high-precision landing theme (≤ 100 m accuracy). The ≥ 87°S landing latitude is as given by the professor; the public ispace and Space Strategy Fund documents do not state a latitude.

03



CHAPTER 03 · MOONWALK 2.0 · REPORT 04

# Assembly methodology

From a list of blocks to the robot’s job list

A planner in Grasshopper reads the block positions from the design model, checks every pick and placement with the Dragon’s real arm, and writes the sequence the robot executes.

03 · ASSEMBLY METHODOLOGY

# From a list of blocks to the robot's job list

The drawings, the animations and the tables are one data set. A step toward 4D BIM: the model gains time, and its time steps are the robot's jobs.




THE PLANNING CHAIN: TWIN → SEQUENCE → REACH → STATIONS → TIMELINE → JOB LIST, WITH EXECUTION AND MEASUREMENT FEEDING BACK INTO THE TWIN

## The rules

Every block is set next to blocks already in place, so each one is supported and the interlock is used from the first course. The robot always works from outside the laid area. On the pad it never drives over set pavers; in the dome it works from the open floor, through a doorway left open until last.

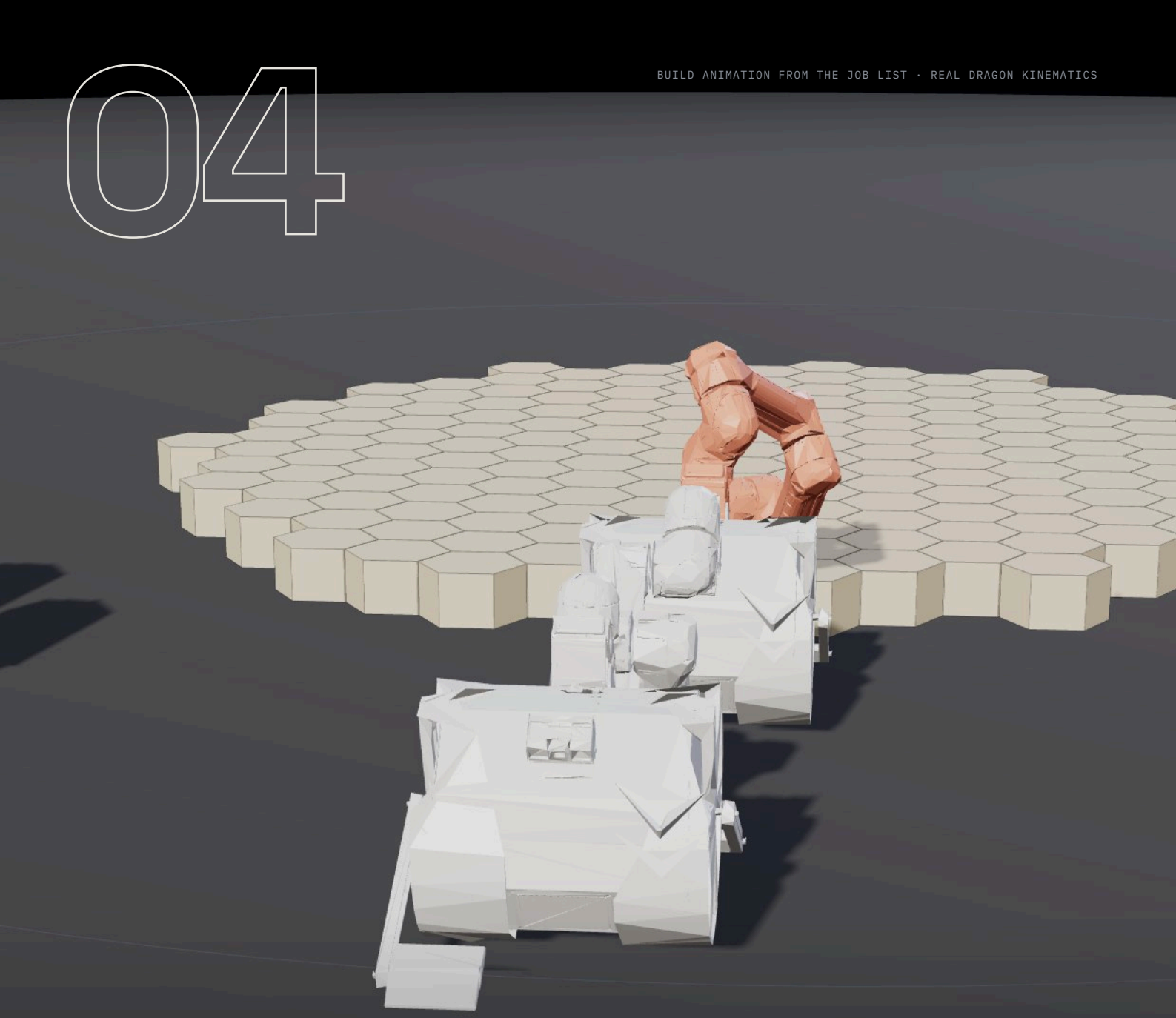
## The checks

A step enters the plan only if the Dragon's real arm, from the lab's robot model, reaches both the pre-place and the place pose. The robot body must also fit: under the shell inside the dome, beside the stack at the stock yard. A block that fails every station is reported, not forced.

## What the robot reads

Each row of the job list holds the step, block ID, ring or course, stock stack and level, the pick station and the place station (x, y, heading), the joint angles, and start and end time. The same file drives the animation, Isaac Sim, and through ROS 2 the robot.

04



CHAPTER 04 · MOONWALK 2.0 · REPORT 04

# Assembly plan · pad

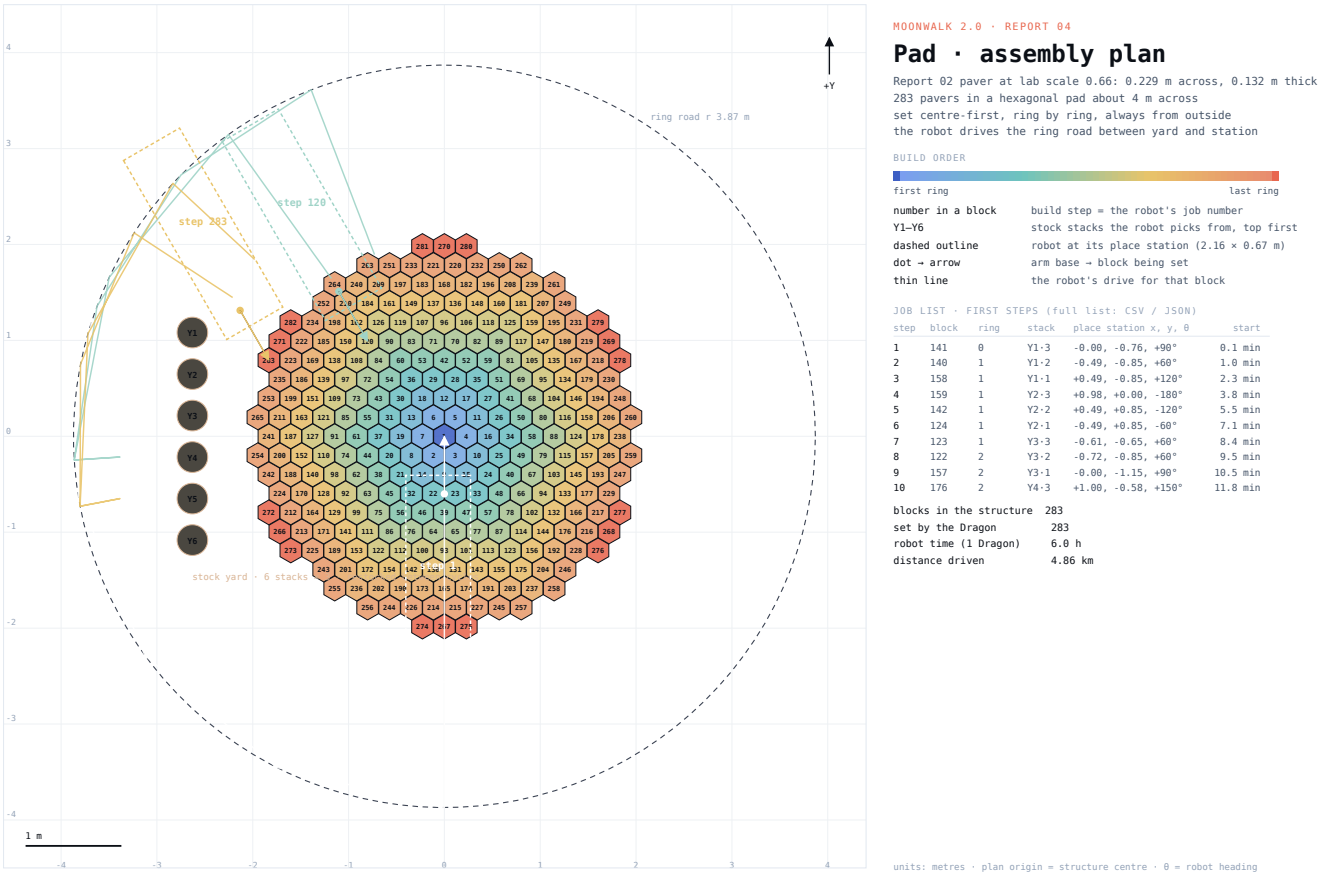
283 pavers, centre out

One Dragon picks each paver from a six-stack yard, drives the ring road to a station outside the laid area, and sets the paver centre-first, ring by ring. Robot time 6.0 h; distance driven 4.9 km.

04 · ASSEMBLY PLAN · PAD

# Pad: 283 pavers, centre out

The Report 02 paver at the lab scale of 0.66 (0.229 m across, 0.132 m thick, about 14 kg in sintered regolith), laid as a 4 m hexagonal pad.




ASSEMBLY PLAN, PLAN VIEW. NUMBERS ARE BUILD STEPS, THE ROBOT'S JOB NUMBERS. DASHED OUTLINES: THE ROBOT AT ITS PLACE STATION FOR THREE EXAMPLE STEPS, WITH ITS DRIVE FOR THAT STEP.

BLOCKS
**283**
pavers in a 4 m hexagonal pad

ROBOT TIME
**6.0 h**
one Dragon

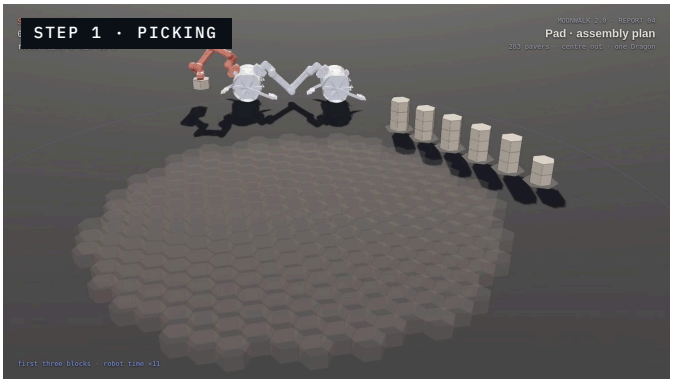
DISTANCE
**4.9 km**
driven on the ring road

RATE
**≈ 47 / h**
pavers an hour, nearly constant

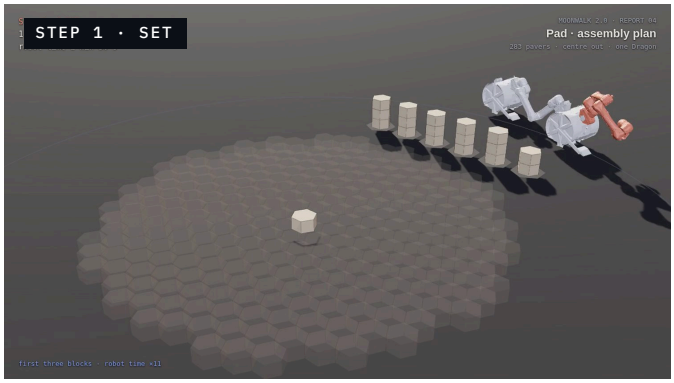
04 · BUILD SEQUENCE

# From the first paver to the last

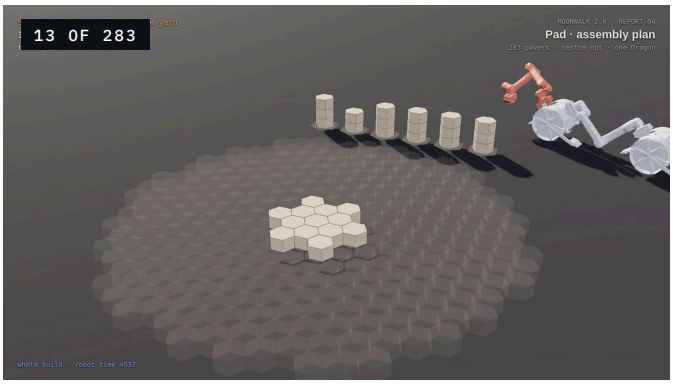
Stills from the one-minute video in the live report, made from the same job list with the Dragon's real kinematics: the first three pavers in detail, then the whole pad as a time-lapse.



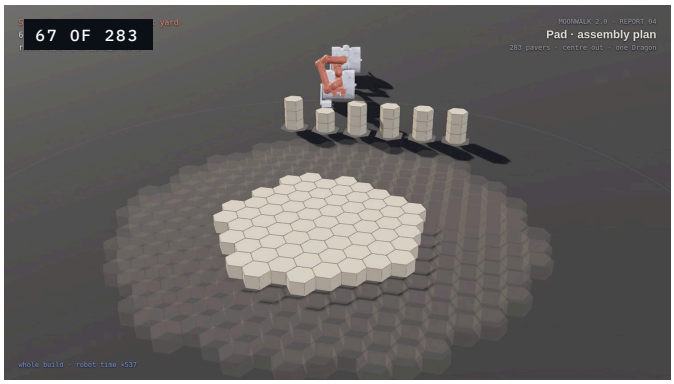

The first paver is lifted from the top of a stack in the yard.



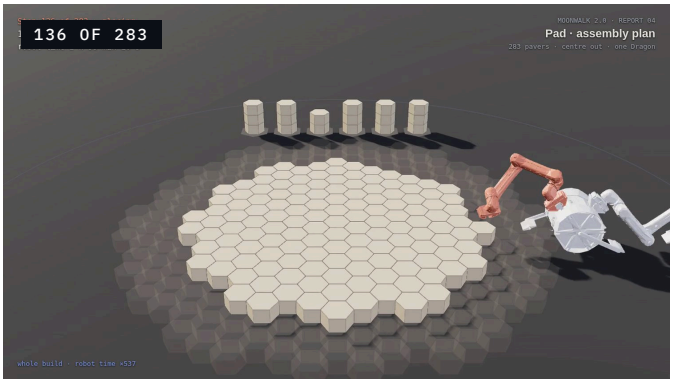

The centre paver is set; the robot backs out to the ring road.



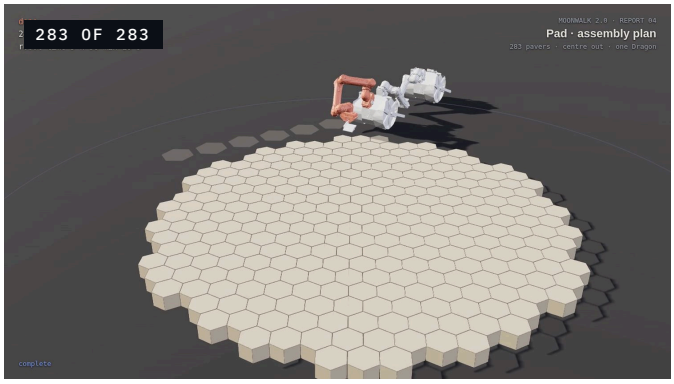

The first ring closes around the centre.




Ring by ring: every paver goes next to pavers already set.

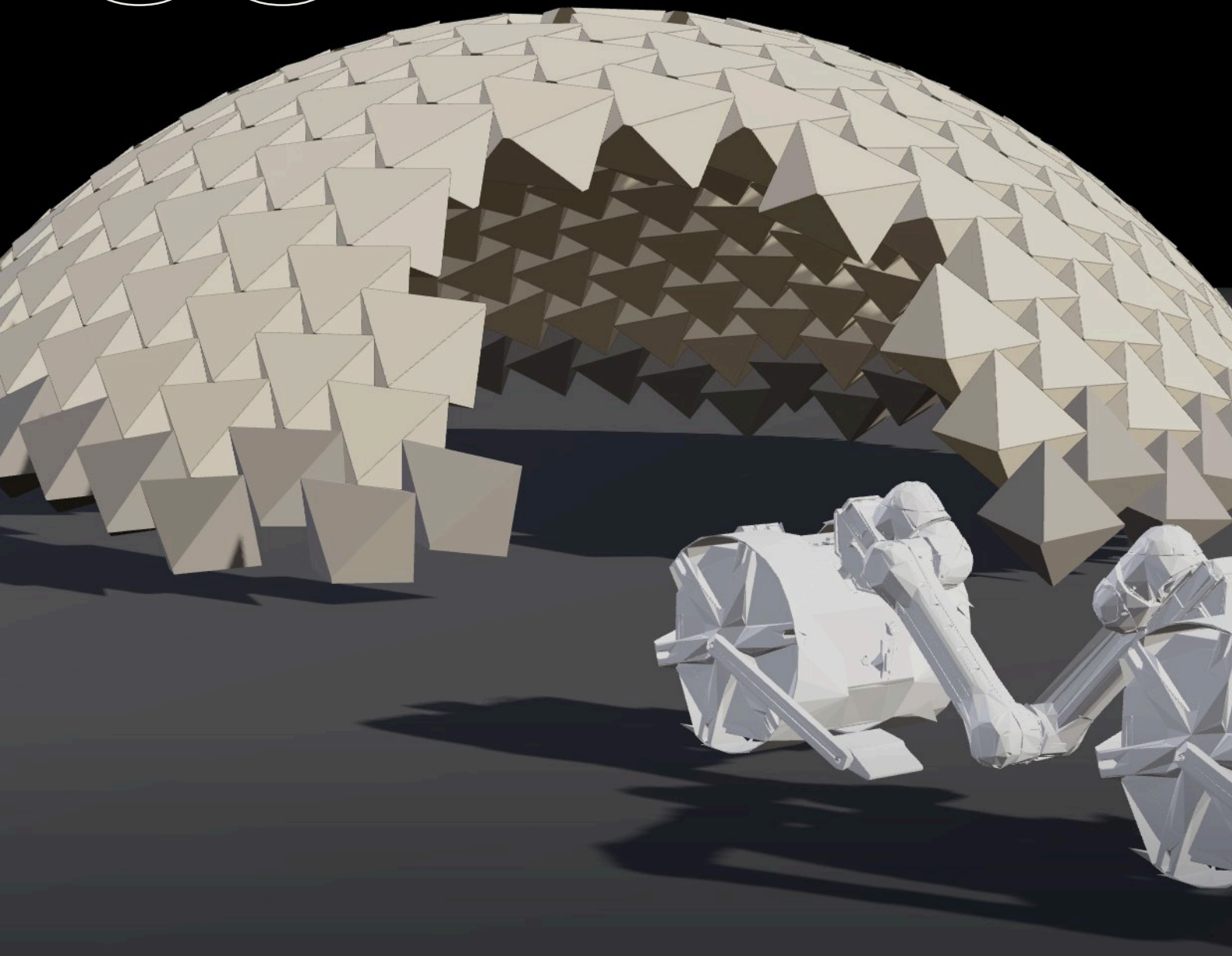



The robot sets from outside the laid area and never drives over it.




Complete: 5 h 58 min of robot time.

05



CHAPTER 05 · MOONWALK 2.0 · REPORT 04

# Assembly plan · dome

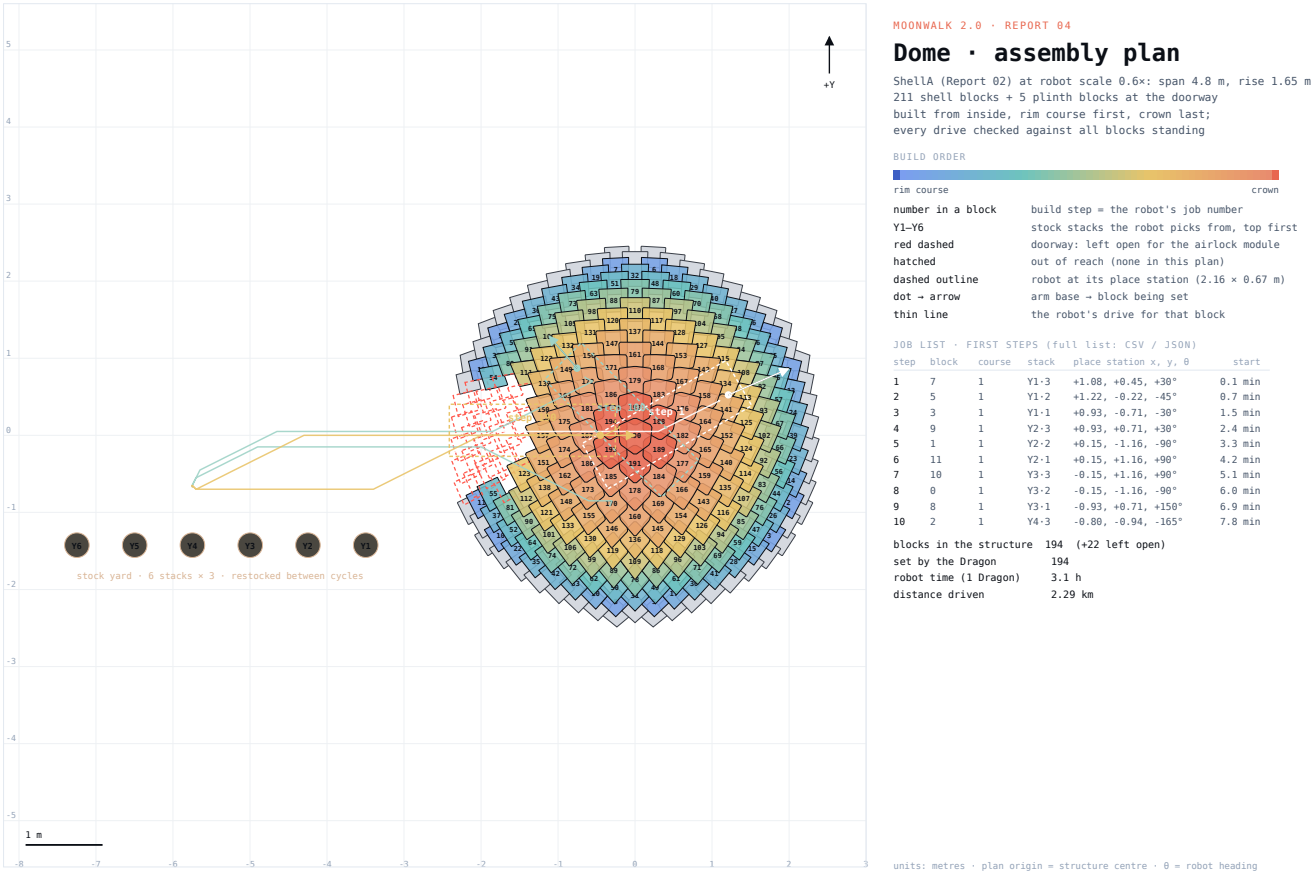
Built from inside, through its own doorway

The Report 02 shell at the size one Dragon can build: 0.6×, 4.8 m across and 1.65 m high. The robot drives in through a doorway for every block, sets the courses rim-first from inside, and drives out at the end.

05 · ASSEMBLY PLAN · DOME

# Dome: built from inside, through its own doorway

ShellA, 211 tilt blocks on a 54-block plinth ring, at 0.6×. A doorway 0.95 m wide and 0.98 m high stays open on the stock-yard side: the opening where the airlock plug-in from Report 03 docks. All 194 blocks around the doorway are set; robot time 3.1 h.

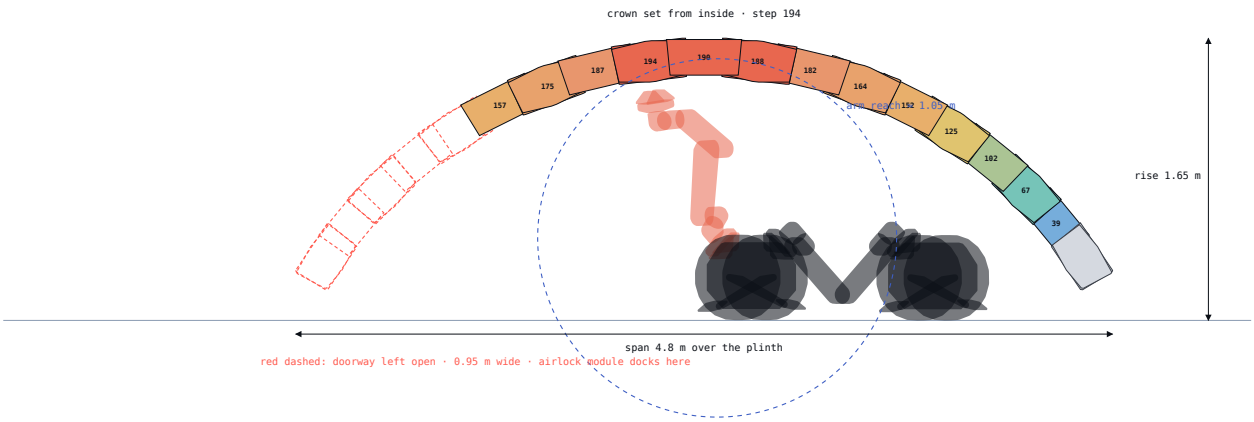



ASSEMBLY PLAN, PLAN VIEW. COURSE COLOURS RUN FROM THE RIM (BLUE) TO THE CROWN (ORANGE). RED DASHED: THE DOORWAY, LEFT OPEN. THIN LINES: THE ROBOT'S COLLISION-FREE DRIVE FOR THREE EXAMPLE STEPS.

MOONWALK 2.0 · REPORT 04

**Dome · section through the doorway**

cut y = 0 ± 0.24 m · numbers = build step · the Dragon sets course 15 from inside (step 194)




SECTION THROUGH THE DOORWAY. THE DRAGON STANDS INSIDE ON THE OPEN FLOOR AND REACHES THE UPPER COURSES FROM BELOW.

## No contact with set blocks

Every pose of the robot is checked against everything standing at that moment: the plinth, every block already set, and the stock stacks, for the body, wheels, legs, arm and the block it carries. Drives are searched on a 5 cm, 15° grid, every turn checked in 5° steps. An independent check over 111,886 drive frames finds no overlap deeper than 4 cm, within the 5 cm grid; these 21 frames are mostly the first moments of backing away from a block just set.

## Why 0.6× and a doorway

The Dragon is 2.2 m long and must stand under the shell. At 0.43× it fits under only the lower courses; at 0.6× it reaches the crown from inside. The doorway has to clear the robot's height in its driving posture (0.86 m), so it cuts through the lower courses on one side: 22 blocks are left out. Closing it from outside does not work, so the opening becomes the entrance.

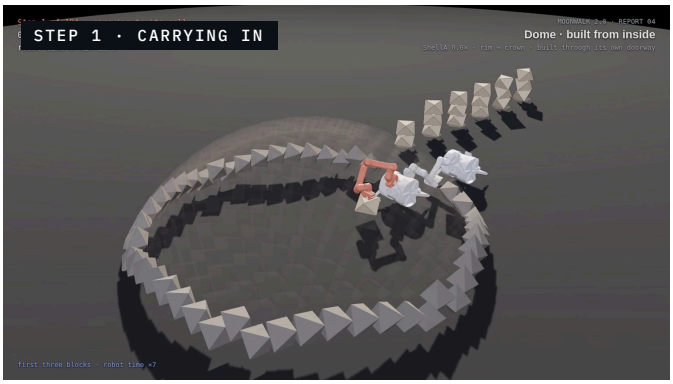
## What is still open

The courses over the doorway span it as a short arch and need a temporary frame while they close; Bruun et al. (2021) used a second robot as that support. At 0.6× a block weighs 30–51 kg in sintered regolith, above the ≈ 12 kg per limb expected on the Moon: lighter, hollow blocks or two limbs per block close that gap. The arm's approach is checked by reach, not yet for contact with neighbouring blocks.

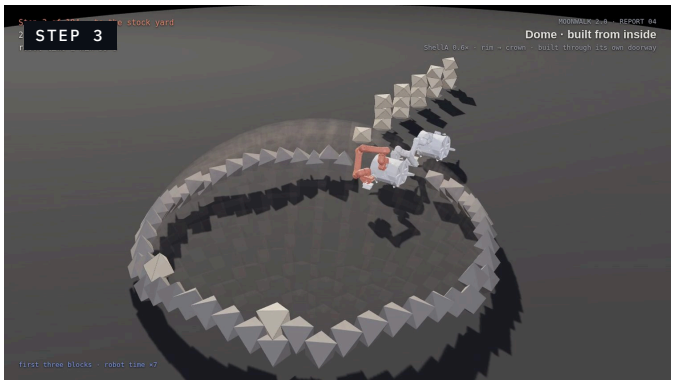
05 · BUILD SEQUENCE

# Rim first, crown last

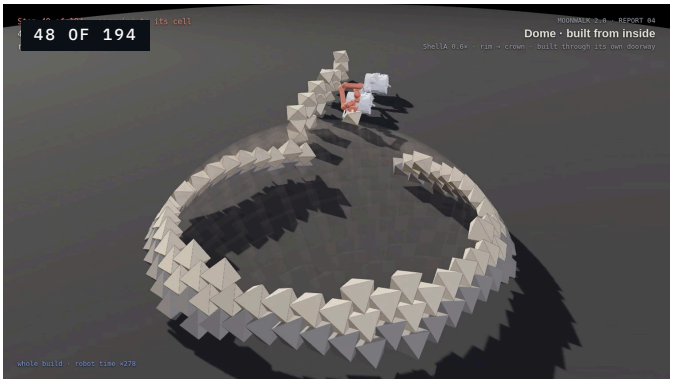
Stills from the one-minute video in the live report: the first three blocks in detail, then the whole dome as a time-lapse, every drive through the doorway.



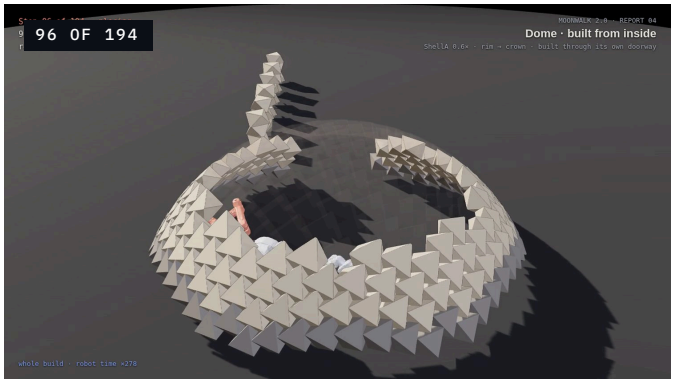

The robot carries the first block in through the doorway.




The rim course starts from the plinth ring.



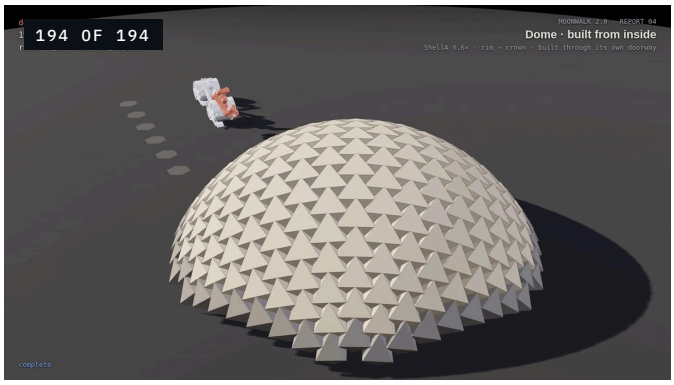

Courses close inwards; the robot works from the open floor.




Every block costs one drive in through the doorway and one out.

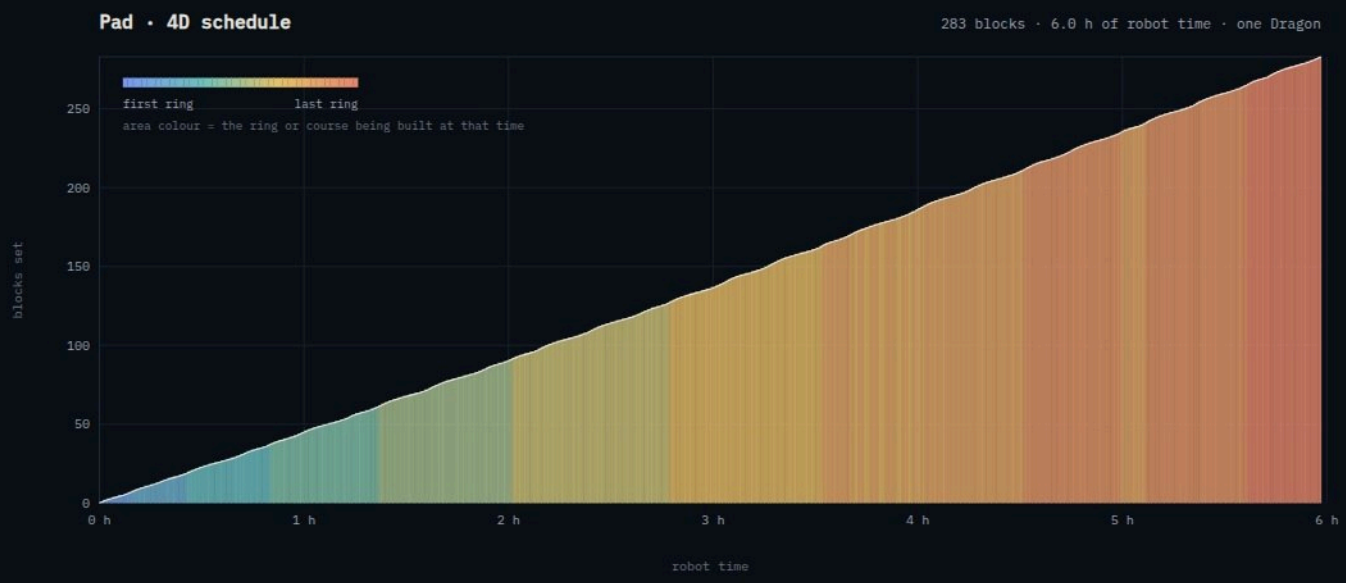



The upper courses, reached from below.




Complete in 3 h 5 min; the robot drives out to its park pose.

06




CHAPTER 06 · MOONWALK 2.0 · REPORT 04

# 4D BIM and robot operations

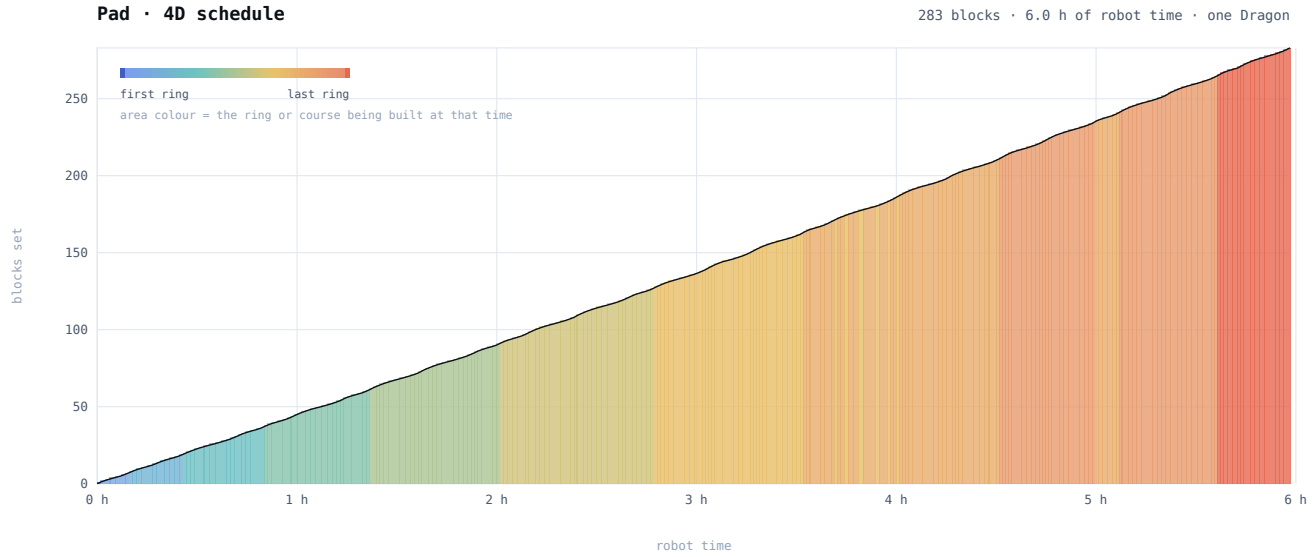
## One model that is also the robot’s schedule

Each step in the job list is a 4D task that carries what the robot needs: which block, which stack, where to stand, which arm posture, and when.

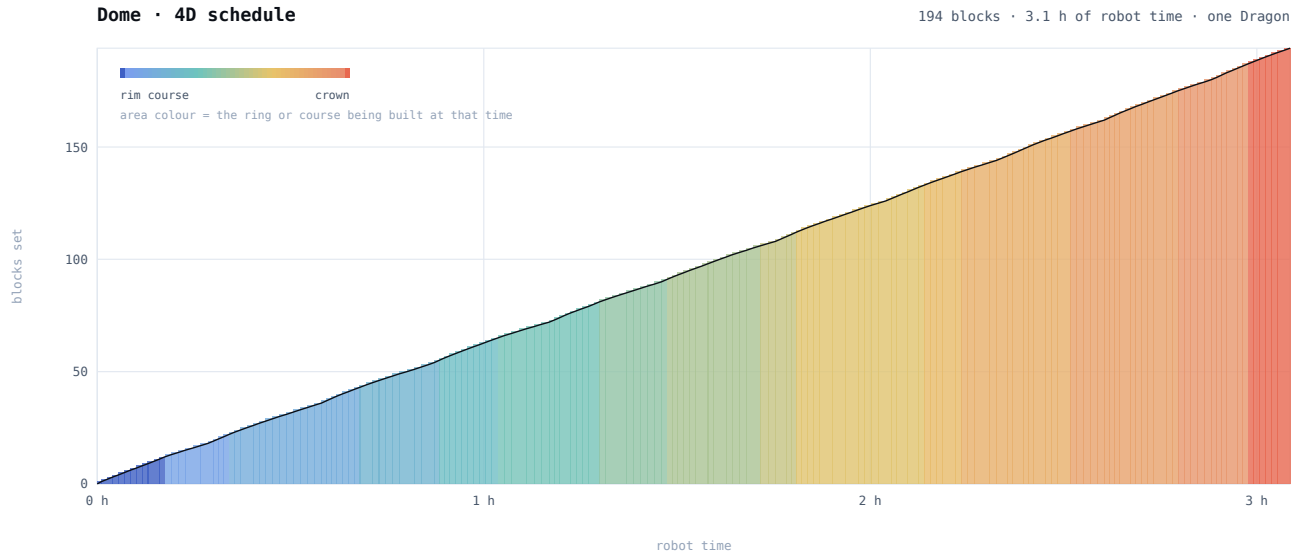
06 · 4D BIM AND ROBOT OPERATIONS

# One model that is also the robot's schedule

The usual 4D BIM links building elements to tasks on a calendar, for people to read. Here the task list is executed by a robot. The animations are those tasks played in time.




PAD: BLOCKS SET AGAINST ROBOT TIME, COLOURED BY THE RING BEING LAID. ABOUT 47 PAVERS AN HOUR, NEARLY CONSTANT, BECAUSE THE RING ROAD KEEPS EVERY DRIVE ABOUT THE SAME LENGTH.




DOME: BLOCKS SET AGAINST ROBOT TIME, COURSE BY COURSE FROM THE RIM TO THE CROWN. ABOUT 63 BLOCKS AN HOUR: EVERY BLOCK COSTS ONE DRIVE IN THROUGH THE DOORWAY AND ONE OUT.

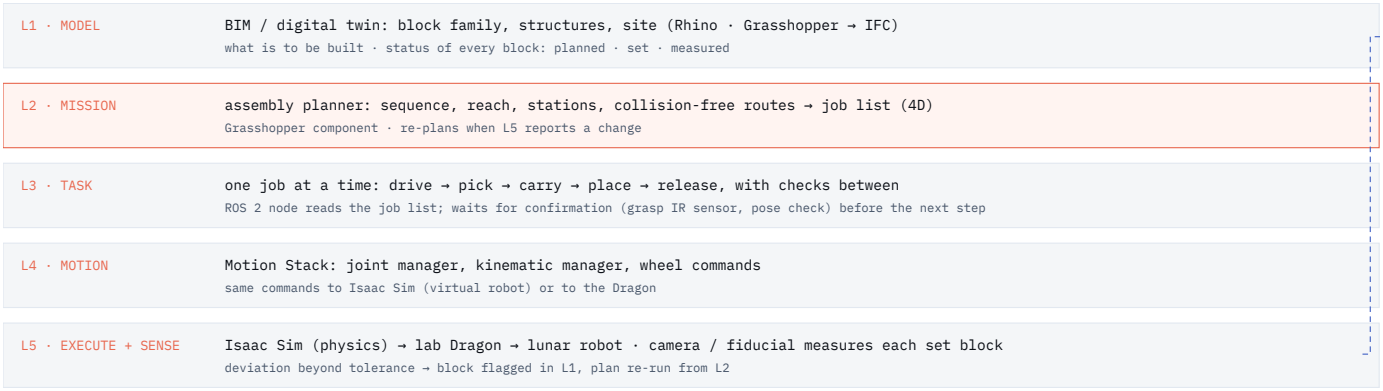
HOW THE PLAN MAPS ONTO IFC

# A proposed IFC 4 mapping

| MOONWALK PLAN | IFC 4 ENTITY | CARRIES |
|---|---|---|
| **block** | IfcBuildingElementPart, in an IfcElementAssembly (pad, dome) | geometry, block type, mass; property set with ring or course, zone, build step |
| **job (one row of the job list)** | IfcTask + IfcTaskTime | step number, scheduled start and finish, predecessor (the previous step) |
| **job ↔ block** | IfcRelAssignsToProduct | which block a job sets |
| **Dragon** | IfcConstructionEquipmentResource | robot ID, assigned to every job (IfcRelAssignsToResource) |
| **stock stack, station** | IfcAnnotation / IfcVirtualElement with placement | pick and place poses (x, y, heading), stack and level |
| **arm posture** | property set on the task | 7 joint angles at pre-place and place; full timeline in JSON beside the model |

A proposed mapping, not yet exported: the planner writes CSV and JSON today. Writing the same data as IFC (for example with IfcOpenShell) lets any BIM viewer play the 4D schedule and lets the site twin record each block as planned, set, or measured.

THE ROBOT OPERATIONS FRAMEWORK




FIVE LAYERS: L1 MODEL, L2 MISSION, L3 TASK, L4 MOTION, L5 EXECUTE AND SENSE; DEVIATIONS GO BACK TO L1 AND THE PLAN IS RE-RUN FROM L2

## What changes from 4D BIM

A 4D BIM schedule is usually carried out by a crew who adapt on site. A robot does not adapt unless told to, so every task must already be feasible: reachable, collision-free, and with the block supported by blocks set before it. The planner proves this for every step.

## Dry stacking as a rule

With no joints to hold a block, the order is part of the structure. The pad goes centre-out; the dome goes course by course, so every block rests on the course below. The arch over the doorway needs a temporary support while its courses close over it.

## Closing the loop

The twin keeps three states per block: planned, set, measured. When a block is measured off its pose, or a step fails in simulation, the block goes back to planned and L2 re-plans the rest from that state. The same loop runs in Isaac Sim today and on the robot later.



CHAPTER 07 · MOONWALK 2.0 · REPORT 04

# Simulation methodology

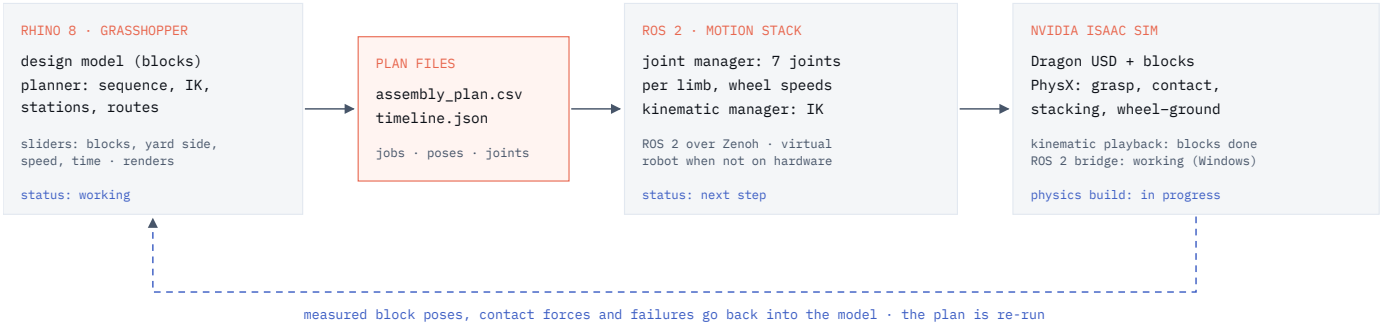
Grasshopper plans, Motion Stack moves, Isaac Sim checks

Each tool does the part it is best at, and the same plan files pass between them. The measured result goes back to the design model.

07 · SIMULATION METHODOLOGY

# Grasshopper plans, Motion Stack moves, Isaac Sim checks

Grasshopper plans the build, Motion Stack (the Space Robotics Lab's ROS 2 software for the MoonBots) turns the plan into joint commands, and Isaac Sim runs those commands with physics.




RHINO 8 · GRASSHOPPER → PLAN FILES → ROS 2 · MOTION STACK → NVIDIA ISAAC SIM, WITH MEASURED POSES, CONTACT FORCES AND FAILURES GOING BACK INTO THE MODEL

## 1 · Plan in Grasshopper

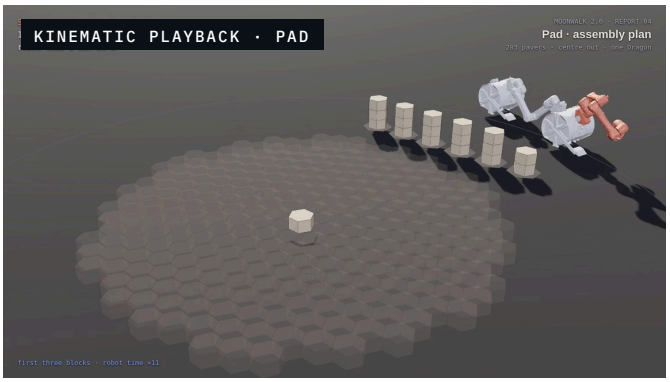
The design model and the planner sit in one Grasshopper definition. A Python component holds the Dragon's kinematics, taken from the lab's robot model and checked against it to within 0.0 mm. Sliders set how many blocks to plan, which side the stock yard is on, and the playback speed. A Bake button writes the job list and the timeline.

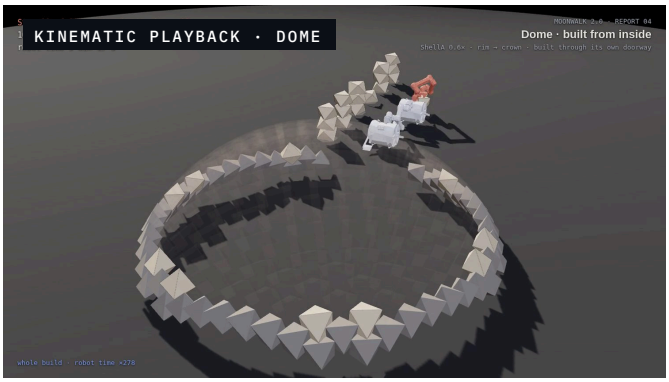
## 2 · Command through Motion Stack

Motion Stack is the open-source ROS 2 software the lab runs on the MoonBots (Neppel et al., 2025). Its joint and kinematic managers take joint targets per limb, and its orchestration swaps in a virtual robot when no hardware is attached. Next step: read timeline.json into Motion Stack, so the same commands drive the simulated Dragon and later the real one.

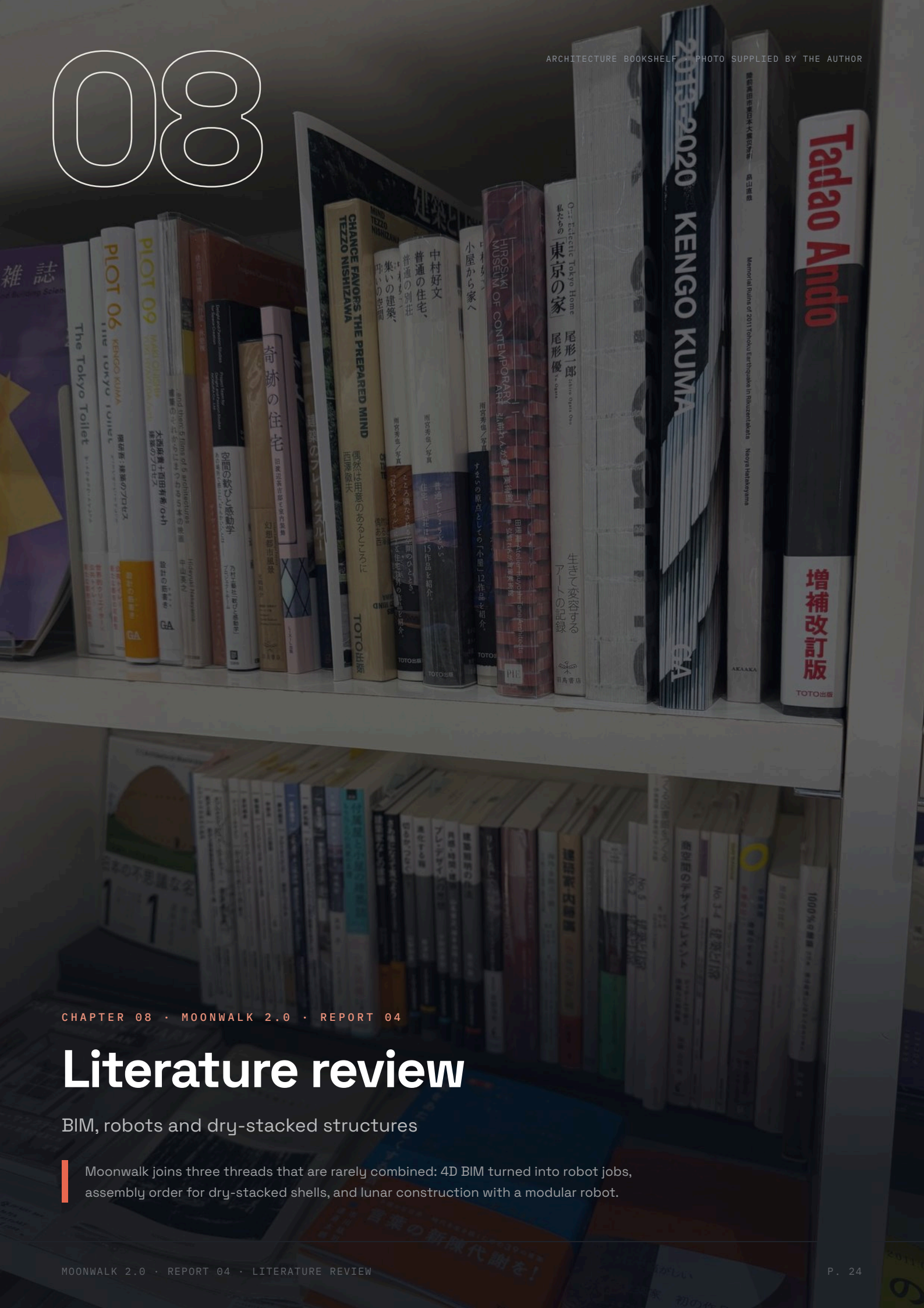
## 3 · Check in Isaac Sim

Isaac Sim runs the Dragon and the blocks with physics: grasp, contact, seating a block, wheels on the ground. The Isaac Sim ↔ ROS 2 link works, and the kinematic playback already moves the blocks; the robot's own playback is still being fixed. The first physics runs tipped the robot and lost blocks from the gripper; the next round fixes the gripper contact and the balance on its wheels.



# 08




ARCHITECTURE BOOKSHELF · PHOTO SUPPLIED BY THE AUTHOR

CHAPTER 08 · MOONWALK 2.0 · REPORT 04

# Literature review

## BIM, robots and dry-stacked structures

Moonwalk joins three threads that are rarely combined: 4D BIM turned into robot jobs, assembly order for dry-stacked shells, and lunar construction with a modular robot.

08 · LITERATURE REVIEW

# Where this work sits

Two recent reviews frame the field. Yue et al. (2025) sort BIM–robot research into task planning, assembly, printing and inspection, linked in four layers. Odugu et al. (2025) find that most systems still pass BIM to the robot one way, with little real-time feedback.

## 1 · REVIEWS AND FRAMEWORKS

FRAMEWORK · DATA LOOP

doi.org/10.1016/j.compind.2025.104299

**Yue, Wang, Zhao, Lai, Huang (2025)**

Interactions between BIM and robotics: towards intelligent construction engineering and management. *Computers in Industry* 169, 104299.

INTEGRATION METHODS

doi.org/10.26599/JIC.2025.9180103

**Odugu, Ghafari, Shourangiz, Khan, Wang (2025)**

BIM-driven robotics in construction: a systematic review of integration methods, applications and future directions. *Journal of Intelligent Construction* 3(4).

BIM FOR ROBOTS (BIMFAR)

doi.org/10.1016/j.compind.2021.103570

**Zhang, Luo, Xu (2022)**

Towards fully BIM-enabled building automation and robotics: a perspective of lifecycle information flow. *Computers in Industry* 135, 103570.

## 2 · FROM BIM AND 4D BIM TO ROBOT TASKS

CLOSEST TO THIS PIPELINE: 4D BIM → ROS → SIMULATION

doi.org/10.36680/j.itcon.2025.015

**Oyediran, Turner, Kim, Barrows (2025)**

Integration of 4D BIM and robot task planning: action-level simulation of indoor wall frame installation. *ITcon* 30, 352–374.

TASK PLANNING · SIMULATION

doi.org/10.1016/j.autcon.2021.103720

**Kim, Peavy, Huang, Kim (2021)**

Development of BIM-integrated construction robot task planning and simulation system. *Automation in Construction* 127, 103720.

WORLD MODEL FROM BIM

doi.org/10.1016/j.autcon.2022.104247

**Kim, Peavy (2022)**

BIM-based semantic building world modeling for robot task planning and execution. *Automation in Construction* 138, 104247.

IFC → ROS · BRICKLAYING

doi.org/10.15480/882.13543

**Brinkhoff, Kolani (2024)**

Deriving fabrication information from BIM for automated robotic task planning. *35th Forum Bauinformatik*, 188–195.

IFC → ROS NAVIGATION

doi.org/10.22260/ISARC2021/0010

**Karimi, Braga, Iordanova, St-Onge (2021)**

Semantic optimal robot navigation using building information on construction sites. *ISARC 2021*, 57–64.

## 3 · DIGITAL TWINS, ROS AND SIMULATION

AS-BUILT WRITTEN BACK TO BIM

doi.org/10.1016/j.compind.2024.104112

**Wang, Yu, McGee, Menassa, Kamat (2024)**

Enabling BIM-driven human-robot collaborative construction workflows with closed-loop digital twins. *Computers in Industry* 161, 104112.

SIM ↔ ROBOT SYNC

www.iaarc.org/publications/2020_proceedings_of_the_37th_isarc/bi_directional_communication_bridge_for_state_synchronization_between_digital_twin_simulations_and_physical_construction_robots.html

**Liang, McGee, Menassa, Kamat (2020)**

Bi-directional communication bridge for state synchronization between digital twin simulations and physical construction robots. *ISARC 2020*.

STABILITY CHECKED AT EVERY STEP

doi.org/10.1016/j.autcon.2024.105935

**Xu, Zhu, Xu, Shao, Zhang, Zhang (2025)**

FEM-based real-time task planning for robotic construction simulation. *Automation in Construction* 170, 105935.

ISAAC SIM LUNAR TERRAIN · SAME LAB

arxiv.org/abs/2309.08997

**Richard, Kamohara, Uno, Santra, van der Meer, Olivares-Mendez, Yoshida (2024)**

OmniLRS: a photorealistic simulator for lunar robotics. *IEEE ICRA 2024*.

MOTION STACK · ROS 2

arxiv.org/abs/2511.01437

**Neppel, Karimov, Mishra, Diaz, Gozbasi, Santra, Uno, Yoshida (2025)**

Designing for distributed heterogeneous modularity: software architecture and deployment of MoonBots.

## 4 · ASSEMBLY ORDER FOR DRY-STACKED SHELLS AND BLOCKS

VAULT BUILT BY ROBOTS, NO FALSEWORK

oar.princeton.edu/bitstream/88435/pr1s17ss4d/1/parascho%20et%20al.%20-%202021%20-%20lightvault%20a%20design%20and%20robotic%20fabrication%20method%20for%20complex%20masonry%20structures.pdf

**Parascho, Han, Beghini, Miki, Walker, Bruun, Adriaenssens (2020)**

LightVault: a design and robotic fabrication method for complex masonry structures. *Advances in Architectural Geometry 2020*.

ROBOTS AS TEMPORARY SUPPORTS · FITS THE DOORWAY ARCH

doi.org/10.1016/j.autcon.2021.103803

**Bruun, Pastrana, Paris, Beghini, Pizzigoni, Parascho, Adriaenssens (2021)**

Three cooperative robotic fabrication methods for the scaffold-free construction of a masonry arch. *Automation in Construction* 129, 103803.

LESS FALSEWORK BY DESIGN

structurae.info/litterature/papier-de-conference/assembly-aware-design-of-masonry-shell-structures-a-computational-approach

**Kao, Körner, Sonntag, Nguyen, Menges, Knippers (2017)**

Assembly-aware design of masonry shell structures. *IASS 2017*.

ORDER + MOTION PLANNED TOGETHER

arxiv.org/abs/2105.11438

**Huang, Garrett, Ting, Parascho, Mueller (2021)**

Robotic additive construction of bar structures: unified sequence and motion planning. *Construction Robotics*.

RE-PLANNING THE ORDER ONLINE

doi.org/10.1145/3730824

**Wang, Liu, Wang, Vallat, Shi, Parascho, Kamgarpour (2025)**

Learning to assemble with alternative plans. *ACM Transactions on Graphics* 44(4).

DRY STACKING FROM LOCAL MATERIAL

doi.org/10.1126/scirobotics.abp9758

**Johns, Wermelinger, Mascaro, Hutter et al. (2023)**

A framework for robotic excavation and dry stone construction using on-site materials. *Science Robotics* 8(84).

SEQUENCING FOR SPACE STRUCTURES

arxiv.org/abs/2310.17115

**Nagpal, Mehr (2023)**

Optimal robotic assembly sequence planning (ORASP).

## 5 · LUNAR CONSTRUCTION AND ISRU BLOCKS

THE ROBOT OF THIS WORK

doi.org/10.1109/TFR.2025.3624346

**Uno, Neppel, Diaz et al., Yoshida (2025)**

MoonBot: modular and on-demand reconfigurable robot toward Moon base construction. *IEEE Transactions on Field Robotics*.

MORTAR-FREE INTERLOCKING

doi.org/10.1016/j.actaastro.2004.12.005

**Dyskin, Estrin, Pasternak, Khor, Kanel-Belov (2005)**

The principle of topological interlocking in extraterrestrial construction. *Acta Astronautica* 57(1), 10–21.

INTERLOCKING PAD PAVERS

technology.nasa.gov/patent/KSC-TOPS-89

**NASA KSC and Sidus Space**

Lunar landing pads: interlocking pavers (KSC-TOPS-89, US Patent 10,920,377).

PADS FIRST, ROBOT-BUILT

ntrs.nasa.gov/citations/20220013524

**Clinton, Edmunson, Effinger et al. (2022)**

NASA's Moon-to-Mars Planetary Autonomous Construction Technology project. *IAC 2022*.

ASSEMBLY ORDER · NASA ARMADAS

doi.org/10.1109/ICRA55743.2025.11127644

**Peters, Cheung, Kostitsyna (2025)**

Assembly order planning for modular structures by autonomous multi-robot systems. *IEEE ICRA 2025*.

SINTERED REGOLITH BLOCKS

iafastro.directory/iac/archive/browse/IAC-18/E5/1/47746/

**Imhof, Sperl, Weiss et al. (2018)**

Using solar sintering to build infrastructure on the Moon: RegoLight. *IAC 2018*.

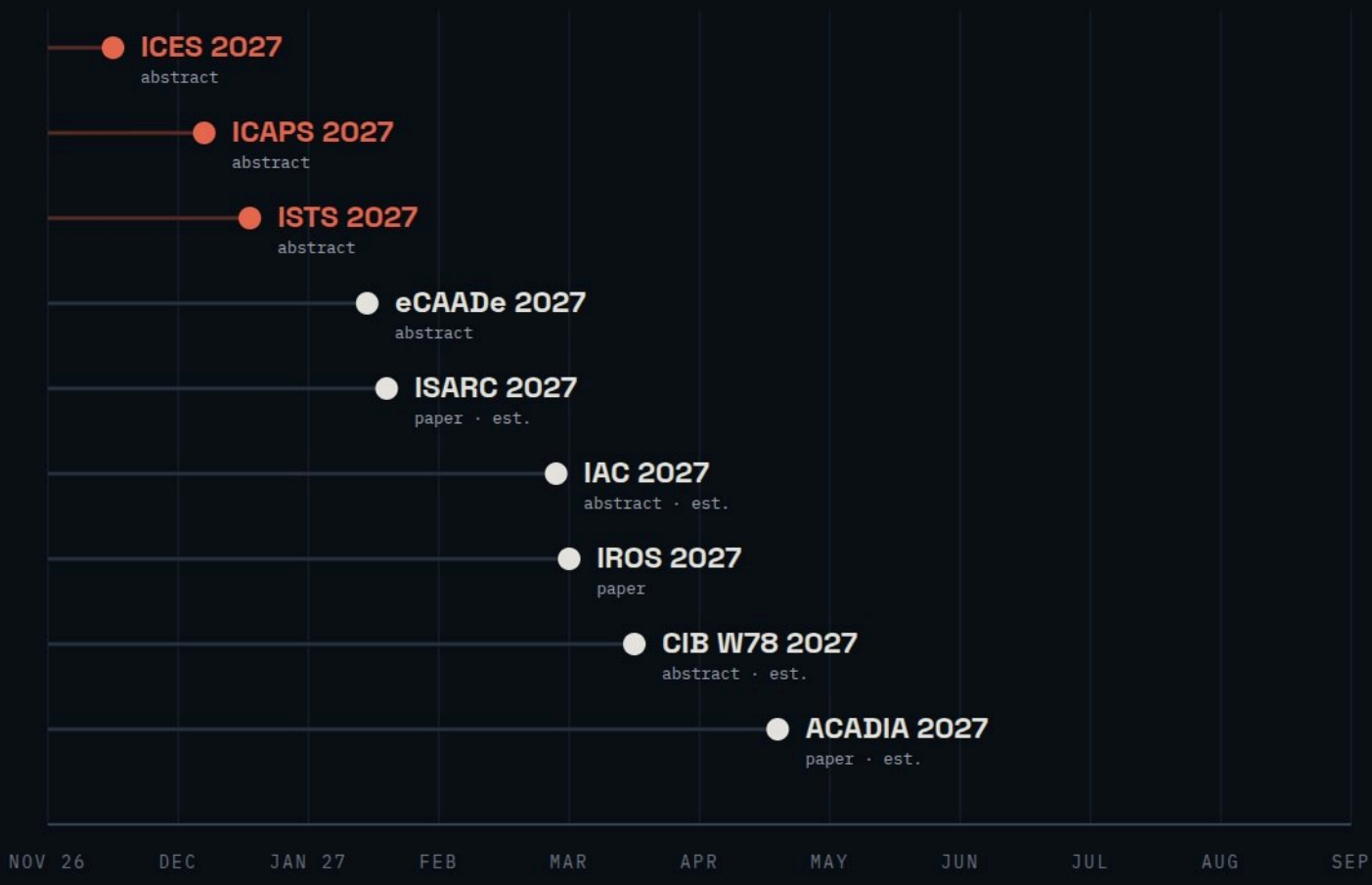
THE GAP THIS WORK FILLS

### Three threads, one modular robot

BIM-to-robot work so far is mostly indoor and on Earth: walls, drywall, bricklaying, with industrial or mobile manipulators. Dry-stacked shell research plans the order and the support, but for fixed arms beside the structure. Lunar construction studies set the goals, pads first and then shelters, with little detail on how a robot sequences discrete blocks. Moonwalk combines the three for one modular robot: a 4D BIM model becomes a job list with an order that respects dry stacking, collision-free routes for a mobile robot working inside the structure it builds, and an execution and feedback path through Motion Stack and Isaac Sim.

09

SUBMISSION DEADLINES · NOVEMBER 2026 → SEPTEMBER 2027




CHAPTER 09 · MOONWALK 2.0 · REPORT 04

# Where to present

## Conferences and deadlines

Open calls after 19 October 2026, nearest first. Dates marked “est.” are estimated from the previous edition and need checking when the call is published.

09 · WHERE TO PRESENT

# Conferences and deadlines

Open calls after 19 October 2026, nearest first. Dates marked “est.” are estimated from the previous edition and need checking when the call is published.

| DEADLINE | CONFERENCE | WHEN AND WHERE | FIT FOR THIS WORK |
|---|---|---|---|
| SOON **16 Nov 2026 · abstract** | **ICES 2027** | 18–22 Jul 2027 · St. Louis, USA | Space architecture (ICES502), regolith ISRU (ICES308) |
| SOON **7 Dec 2026 · abstract**<br>**14 Dec · paper** | **ICAPS 2027** | 27 Jun–2 Jul 2027 · Columbia, USA | Assembly and 4D planning as a planning problem; PlanRob workshop (est. Mar 2027) |
| SOON **17 Dec 2026 · abstract** | **ISTS 2027** | 17–23 Jul 2027 · Toyama, Japan | Japan’s main space symposium; natural venue with the Tohoku lab |
| **15 Jan 2027 · abstract**<br>**1 Apr · full paper** | **eCAADe 2027** | 23–27 Aug 2027 · Espoo, Finland | Computational design, Grasshopper, 4D BIM |
| **~ mid/late Jan 2027 · paper (est.)** | **ISARC 2027** | summer 2027 · Odense, Denmark | Construction robotics, BIM, digital twins: the best fit for the method |
| **~ late Feb / early Mar 2027 · abstract (est.)** | **IAC 2027** | 27 Sep–1 Oct 2027 · Poznań, Poland | Lunar infrastructure, space architecture (E5), building blocks (D3) |
| **1 Mar 2027 · paper** | **IEEE IROS 2027** | 26 Sep–1 Oct 2027 · Florence, Italy | Modular robots, assembly, simulation; with the lab |
| **~ Mar 2027 · abstract (est.)** | **CIB W78 2027** | 2027, host to be announced | BIM and IT in construction |
| **~ Apr–May 2027 · paper (est.)** | **ACADIA 2027** | 2027, host to be announced | Architecture and robotics |
| **call not yet published** | **Space Resources Week 2027** | 10–13 May 2027 · Luxembourg | ISRU and lunar construction industry; VR demo and model |
| **~ mid 2027 · for 2028 (est.)** | **ASCE Earth & Space 2028** | 2028, every two years | Lunar construction, regolith, robotics |
| **~ Sep 2027 · for 2028 (est.)** | **Rob\|Arch 2028** | 2028, every two years | Robots in architecture |

Already closed for 2027: ICRA 2027 (papers closed 16 Sep 2026), CAADRIA 2027, CAAD Futures 2027, IEEE Aerospace 2027, AIAA ASCEND 2027. Worth attending this year without a paper: i-SAIRAS + iSpaRo, 3–6 Nov 2026, Cologne.

SOURCES

Neppel et al., Designing for Distributed Heterogeneous Modularity: On Software Architecture and Deployment of MoonBots (arXiv 2511.01437; Motion Stack, github.com/2lian/Motion-Stack) · Uno et al., MoonBot, IEEE Transactions on Field Robotics, 2025, DOI 10.1109/TFR.2025.3624346 · candidate regions and crater positions: Report 02 (NASA Artemis III regions, 2022 and 2024 lists; positions approximate). Robot: MoonBot Dragon (dragon_v2), kinematics from the lab model · assembly plans: Grasshopper planner (moonwalk_gh).



The SPACE ROBOTICS LAB

# Moonwalk 2.0

REPORT 04 · FROM THE PLAN TO THE ROBOT

LIVE REPORT
arhimatrix.github.io/arhimatrix-moonwalk-2-rep_04

INTERACTIVE PARTS
3D Moon tour with the 87° line · 3D build players for the pad and the dome · videos · job lists (CSV, JSON)

REPORTS
Report 01 · 02 · 03 · 04 at arhimatrix.github.io

FILES
pad and dome job lists (CSV) and timelines (JSON) with the live report

NINA VELIMIROVIĆ · SPACE ROBOTICS LABORATORY, TOHOKU UNIVERSITY · OCTOBER 2026